

helix_proxy — Feature Status

Revision: 3 **Last modified:** 2026-07-01T15:30:00Z **Status:** Data plane (Squid HTTP/HTTPS forward + Dante SOCKS5 + Squid caching) is deployed, end-user-reachable, and PASS with captured evidence. The VPN-aware control-plane (dynamic routing / control API / health publisher / breaker / ACL-helper / PAC / metrics) is a separate Go module — component-tested (unit/integration GREEN + §1.1 mutations) but NOT yet wired into the running data plane, therefore PENDING at the end-user layer. Let's Encrypt HTTPS: Phase-1 cert-analyzer PASS **and** Phase-2/3 hermetic DNS-01 issuance now **PASS** (re-runnable end-to-end via a custom DNS-module-linked Caddy image + Pebble, live Challenge + standing regression guard, cert chains to the per-run Pebble CA); Phase-5 renewal/rotation is **PENDING** (diagnosed — ARI on Pebble 2.6.0 blocks in-process renewal; needs Pebble $\geq 2.8.0$ /set-renewal-info/); staging/production cutover (Phase 4/6) OPERATOR-BLOCKED. **Authority:** Inherits constitution/Constitution.md per §11.4.35. This is the §11.4.153 per-feature Status set (companion §11.4.56 two-audience Status_Summary.md). Every row is reconciled against real code + committed tests/challenges/evidence; no metadata-only PASS (§11.4 / §11.4.1 / §11.4.6). **Repo anchor:** branch feature/vpn-aware-dynamic-routing, HEAD 2f1e49d at authoring time (LE issuance b2afa7d, Challenge 022b78d, guard d2042ac, renewal research 6349c0e+7c6b69f+2f1e49d).

Operator-blocked / operator-gated items (read first — §11.4.45 O(1) surface)

Item	Why blocked	Unblock condition
Real VPN egress proof (egress-IP $\neq$ host-IP through the proxy)	No live tunnel bound at test time; proxy-vpn container observed Stopped ; VPN_EXIT_IP unset. The verify scripts SKIP honestly (operator_attended) rather than fabricate a VPN PASS (tests/	Operator brings up a real tunnel and exports VPN_EXIT_IP=<tunnel exit>; then tests/egress_proof/real_vpn_egress_proof.sh produces the live RED→GREEN proof.

Item	Why blocked	Unblock condition
Let's Encrypt Phase 4 — LE staging end-to-end (real DNS-01)	verify-proxy.sh:87-98). Needs a real DNS provider + a scoped API token stored as a Podman secret (§11.4.10 — value never in git/.env). Operator chose <i>defer real domain</i> (design §9 Phase 0).	Operator provides the DNS provider name + a scoped API token + a test hostname.
Let's Encrypt Phase 6 — production cutover	Real public domain + real DNS-01 for that domain + a single operator-gated go-live (design §9).	Operator authorises go-live with the real domain + token + host firewall opening.

Feature status table (§11.4.153)

Legend. Wiring = *is the feature genuinely reachable by an end user of the DEPLOYED stack (docker-compose.yml)?* Validation ∈ {PASS, FAIL, SKIP, PENDING, OPERATOR-BLOCKED}. PASS requires a committed captured-evidence path (§11.4.69). PENDING = code/tests exist but the feature is not yet end-user-reachable in the deployed stack (no metadata-only PASS). Video-confirmation is **N/A** for every row: helix_proxy is a headless network service with no rendered UI — its §11.4.107/§11.4.69 proof is captured curl status codes + Squid Via: headers + access.log TCP_*HIT facts, not a video (§11.4.153(3) autonomous-infeasible ⇒ honest non-video evidence).

A. Data-plane proxy features — DEPLOYED & end-user-reachable

Component	Feature	Category	Implementation (file:line)	Wiring (end-user-reachable?)
Squid :53128	HTTP forward proxy (plain-HTTP GET relayed to upstream)	functional / data-plane	config/squid/squid.conf:53 (http_access allow localnet); docker-compose.yml:56 (proxy-squid)	YES — curl -x http://localhost:53128 forwards live
Squid :53128				

Component	Feature	Category	Implementation (file:line)	Wiring (end-user-reachable?)
Dante :51080	HTTPS CONNECT tunneling (CONNECT to TLS upstream)	functional / data-plane	config/squid/squid.conf:46,50 (acl CONNECT / deny CONNECT ! SSL_ports); docker-compose.yml:56	YES — curl -x http://localhost:53128 https://...
	SOCKS5 proxy (HTTP over SOCKS5)	functional / data-plane	config/dante/sockd.conf; docker-compose.yml:127 (proxy-dante)	YES — curl --proxy socks5://localhost:51080
	SOCKS5-over-HTTPS (TLS carried over SOCKS5)	functional / data-plane	config/dante/sockd.conf; docker-compose.yml:127	YES — curl --proxy socks5://https://...
Squid :53128	Response caching (cache-HIT of cacheable object)	performance / data-plane	config/squid/squid.conf:9 (cache_dir aufs ... 51200), :10 (cache_mem 512 MB), :70-73 (refresh_pattern)	YES — repeat GET served from cache
Squid :53128	Cache admin/inspection (cachectl stats/list/size)	operational	cachectl (repo root)	YES — operator CLI against live cache
Squid :53128	ACL access control (allow localnet, deny-all default, Safe_ports, CONNECT→SSL-ports-only, manager deny)	security / data-plane	config/squid/squid.conf:49-55	YES — enforced by running Squid on every request

Component	Feature	Category	Implementation (file:line)	Wiring (end-user-reachable?)
	— allow-path			

A-note (anti-bluff, §11.4.6): the ACL *allow* path is proven — every forward/SOCKS5 challenge succeeds only because `http_access allow localnet` admits it. The ACL *deny* paths (`deny all`, `deny !Safe_ports`, `deny CONNECT !SSL_ports`, `deny manager`) are present + enforced by the running Squid but have **no dedicated committed deny-assertion capture** yet, so a full deny-path PASS is deferred to §B (PENDING). Proxy *authentication* (`username/password 407`) is NOT in the deployed `squid.conf` — see §B.

B. Control-plane & auth features — code/tests exist, NOT yet wired into the deployed data plane (PENDING)

The control-plane is a separate Go module (`control-plane/`, module `digital.vasic.helixproxy/controlplane`) that sits *beside* the byte path. Its packages are phased (P3–P6 landed with unit/integration GREEN + §1.1 mutations), but the deployed admin surface is still the `traefik/whoami` placeholder (`docker-compose.yml:199-201`), so none of these are end-user-reachable in the running stack yet. Per §11.4 / §11.4.1 they are **PENDING**, never PASS.

Component	Feature	Category	Implementation (file:line)	Wiring (end-user-reachable?)
control-plane internal/ routing	VPN-aware dynamic routing (render Squid include + Dante routes + PAC)	data-plane routing	control-plane/ internal/routing/ routing.go (+ testdata/*.golden)	NO — dyn stack (doc compose.dy not deploy
control-plane cmd/api + internal/api	Control API (REST CRUD / SSE status / PAC endpoint / / metrics / mTLS)	control API	control-plane/ internal/api/ server.go, handlers.go, metrics.go, tls.go; cmd/api/main.go	NO — adr is traefik/ placeholder
control-plane cmd/healthd + internal/ vpn	VPN health publisher / health endpoints (tunnel-up = tx-	health / observability	control-plane/ internal/vpn/ health.go, gluetun.go, wg.go; cmd/healthd/main.go	NO — not into depla

Component	Feature	Category	Implementation (file:line)	Wiring (e.g. reachable)
	$\Delta > 0 \wedge$ fresh handshake $\wedge$ egress $\neq$ host)			
control-plane internal/ breaker	Per-target circuit breaker + tunnel tier-failover	resilience	control-plane/ internal/breaker/ breaker.go, selection.go, tunnelbreaker.go	NO
control-plane cmd/acl- helper + internal/ aclhelper	External ACL helper (OK tag=<tunnel> / ERR graceful-503, fail-closed)	data-plane ACL	control-plane/ internal/aclhelper/ decide.go, protocol.go; cmd/acl- helper/main.go	NO — Squid external_acl not wired deployed
control-plane internal/pac	PAC generation (FindProxyForURL)	data-plane routing	control-plane/ internal/pac/ generate.go, pac.go (+ testdata/ pac.golden)	NO
Squid :53128	Proxy authentication (username/password 407) + ACL deny-path assertion	security	(design) tests/ dynamic/analyzers/ auth_407_analyzer.sh; not in deployed config/squid/ squid.conf	NO — no auth in deployed
Prometheus / Grafana / OTel	Observability & metrics (scrape topology, dashboards, in-process OTel)	observability	config/prometheus/ prometheus.yml, config/grafana/, control-plane/ internal/otel/ otel.go, internal/ api/metrics.go	NO — prometheus explicitly YET-LIVE scrape target not exist

C. Operational / health surface — DEPLOYED

Component	Feature	Category	Implementation (file:line)	Wiring	Reachable
./status	Operational status/health command (Container/	health / operational	status (repo root)	YES — operator CLI + --json	Reachable via CLI

Component	Feature	Category	Implementation (file:line)	Wiring	R
	Port/ Connection state)				Pe C fi js st he
proxy-admin :58080	Admin container reachability / health (200)	health	docker- compose.yml:199-201 (traefik/whoami)	YES (reachable) — but is the whoami placeholder , not the real admin UI	He fr he
Containers	Compose service healthchecks (Squid/ Dante/admin liveness)	health	docker-compose.yml (healthcheck: blocks)	YES — runtime restart/ liveness	. / re R S pe co

D. Let's Encrypt HTTPS workstream (task #59)

Detailed integration status: [../design/letsencrypt/Status.md](https://github.com/letsencrypt/Status.md).

Component	Feature	Category	Implementation (file:line)	W
cert-analyzer	Offline PEM analyzer (cert_not_expired / cert_days_remaining / cert_san_matches / cert_chain_roots_in / cert_renewal_due) + golden-good/golden- bad self-validation (\$11.4.107(10))	security / TLS	tests/letsencrypt/ cert_analyzer.sh (+ cert_analyzer_selftest.sh, fixtures/)	Y (c to a re c re o o
Custom Caddy image + Pebble	Hermetic DNS-01 ACME issuance (Phase 2/3) — custom caddy- challtestsrv image issues a real leaf from a per-boot Pebble CA over DNS-01, end-to-end	TLS	deploy/letsencrypt/ phase3_hermetic_issue.sh (re- runnable); deploy/ letsencrypt/build.sh + Dockerfile.caddy + caddy- challtestsrv-dns/; deploy/ letsencrypt/Caddyfile	Y (l is st p is c e e p d c

Component	Feature	Category	Implementation (file:line)	Wiring	Real-u
Caddy + Pebble	Renewal / rotation (Phase 5) — in-process cert renewal + zero-downtime hot-swap	TLS	(diagnosed) docs/research/letsencrypt_renewal_20260701/ANALYSIS.md (Rev 3, §11.4.138 live findings)		
LE staging / production	Real DNS-01 issuance (Phase 4) + production cutover (Phase 6)	TLS	(design) docs/design/LETSENCRYPT_HTTPS.md §9		

E. VPN routing / egress

Component	Feature	Category	Implementation (file:line)	Wiring	Real-u
egress proof	Real VPN egress proof (egress-IP through proxy = tunnel exit $\wedge$ $\neq$ host-IP)	data-plane / security	tests/egress_proof/real_vpn_egress_proof.sh; evidence.sh; assert_egress_ip; egress_neq_host analyzer	NO — proxy-vpn observed Stopped; no live tunnel	— (SKIP) operator — the host ip "VPN v bluff w remov

§11.4.169 test-type coverage matrix

Per-feature × test-type × evidence-state. Cells: **PASS** = live captured evidence committed; **UNIT** = unit/fixture-level GREEN (incl. golden-good/golden-bad self-validation) but not a live-stack run; **SKIP** = honest topology SKIP (dynamic/live stack absent — §11.4.3); **PENDING** = warranted but not yet produced; **N/A** = not applicable to this feature class. The DEPLOYED data-plane suites for stress/chaos/ddos/benchmark/memory/concurrency currently **SKIP** because they require the un-deployed dynamic stack (P10) — their analyzers are self-validated and their scripts are parse-clean, so the SKIP is honest, not a gap-hidden PASS.

Feature	unit	integration	e2e / live
HTTP forward proxy	PASS (proxy_conn_verdict_test.sh)	PASS (comprehensive)	PASS (challenge live curl)
HTTPS CONNECT tunneling	PASS	PASS	PASS (challenge)
SOCKS5 proxy	PASS	PASS	PASS
SOCKS5-over-HTTPS	PASS	PASS	PASS
Response caching (TCP_*HIT)	PASS (xcache_hit_analyzer self-val)	PASS (comprehensive TCP_MEM_HIT)	PASS (comprehensive live double-fetch)
Cache admin (cachectl)	PASS (cache_cli_present_test.sh)	PASS (comprehensive)	PASS (live cachectl stats)
Squid ACL (allow-path)	PASS (analyzer)	PASS (live forward)	PASS
Proxy auth (407) / ACL deny-path	UNIT (auth_407 self-val)	PENDING	PENDING
VPN-aware dynamic routing	UNIT (routing golden)	UNIT (integration golden)	SKIP (dynamic stack absent)
Control API (REST/SSE/PAC/mTLS)	UNIT (-race)	UNIT (real Postgres)	PENDING (admin=whoami)
VPN health publisher / health	UNIT	UNIT (real gluetun + chaos)	PENDING
Circuit breaker + tier-failover	UNIT	PENDING	PENDING
ACL helper (graceful 503)	UNIT	UNIT	PASS (live P10 fail-closed)

Feature	unit	integration	e2e / live
PAC generation	UNIT (golden)	UNIT	PENDING
Observability / metrics	UNIT (control /metrics)	PENDING	PENDING (targets absent)
./status operational health	PASS	PASS (comprehensive)	PASS (live)
LE cert-analyzer (Phase 1)	PASS (selftest 37/37 + §1.1)	PASS (regression guard)	N/A (offline tool)
LE hermetic DNS-01 issuance (Phase 2/3)	PASS (cert-analyzer verdicts)	PASS (guard GREEN, wired in run-tests.sh)	PASS (hermetic end-to-end DNS-01 issuance)
LE renewal / rotation (Phase 5)	PENDING	PENDING	PENDING (ARI-blocked on Pebble 2.6.0)
LE staging/production (Phase 4/6)	N/A	N/A	OPERATOR-BLOCKED
Real VPN egress proof	UNIT (egress_neq_host self-val)	SKIP	OPERATOR-BLOCKED (no live tunnel)
ddos / flood resilience	UNIT (ddos_flood_evidence guard)	SKIP (dynamic stack)	SKIP
benchmark / latency ratchet	UNIT (benchmark_baseline_ratchet guard)	SKIP	PASS (live 200×204)

Honest gaps (§11.4.3 / §11.4.6)

1. **HelixQA live execution is SKIP, not PASS.** The helixqa binary cannot be built in this checkout — its go.mod replaces six own-org sibling modules (doc_processor, llm_orchestrator, llm_provider, llms_verifier, vision_engine, security) that are not vendored here. The runner emits an honest §11.4.3 SKIP naming the exact blocker (qa-results/helixqa/20260701T090532Z/SKIP.md). The proxy data plane was independently re-proven

this run via a stdlib replica of the HelixQA HTTPExecutor client (mech_http_forward.txt 204, mech_https_connect.txt 200, mech_socks5.txt 204). Unblock: vendor the six modules under submodules/.

2. **Live stress/chaos/ddos/benchmark/memory/concurrency SKIP.** As of run 20260630T221343Z all six dynamic-suite runs honest-SKIP against the then-undeployed dynamic stack (P10) (qa-results/p9-harness/20260630T221343Z/suite_results.txt: suites=6 PASS=0 SKIP=6 FAIL=0). Their analyzers are self-validated (golden-good PASS / golden-bad FAIL) and their scripts parse-clean; the RED regression guards (ddos_flood_evidence, benchmark_baseline_ratchet) PASS. **Update 2026-07-01:** the dynamic stack was subsequently brought up and two of these now have live PASS this round — the **forward-proxy benchmark** (qa-results/benchmark/proxy_forward_20260701T130414Z/latency.txt: p50=0.086s, 200/200 204) and the separate **P10 VPN fail-closed** dynamic run (qa-results/dynamic/vpn_failclosed/20260701T130115Z/verdict.txt: 3/3 branded 503, 0 leak, graceful_503_rc=0); the control-plane is also race-clean (qa-results/race/control-plane_race_20260701T125739Z.log: 0 DATA RACE, EXIT=0). The remaining stress/chaos/ddos/memory *p9-harness dynamic-stack* suites still SKIP pending their own live runs. (Distinct scope — the **base-proxy** hardening variants of stress+chaos, DDoS, memory-soak, and concurrency DID pass live this session against :53128; see docs/design/hardening/Status.md §11.4.169 matrix. The SKIPS here are the separate dynamic-stack p9-harness suites.)
 3. **ACL deny-path** has enforcement but no dedicated committed deny-assertion capture — represented PENDING, not folded into the allow-path PASS.
 4. **Cache challenge client-side SKIP** (topology_unsupported) is honest — the Squid access.log is owned by a rootless-container subuid and unreadable client-side; the decisive TCP_MEM_HIT PASS comes from comprehensive-test's container-side snapshot instead.
 5. **Video-confirmation N/A** for every row (§11.4.153(3)): helix_proxy is headless — proof is captured status codes + Via: headers + access.log TCP_*HIT, not rendered video.
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Honest boundary (§11.4.6)

Only the **deployed Squid + Dante data plane** (HTTP forward, HTTPS CONNECT, SOCKS5, SOCKS5-over-HTTPS, caching, cache admin, ACL allow-path, ./status, admin reachability), the **LE Phase-1 cert-analyzer**, and the **LE Phase-2/3 hermetic DNS-01 issuance** (re-runnable, live Challenge + standing guard, cert chains to the per-run Pebble CA) are PASS with committed captured evidence. Everything in \$B (control-plane routing / API /

health / breaker / ACL-helper / PAC / metrics, and proxy auth) is **component-tested but not wired into the running stack** — honestly PENDING, never a metadata-only PASS. The LE **renewal/rotation (Phase 5)** path is PENDING — diagnosed, not yet produced (ARI on Pebble 2.6.0 blocks in-process renewal; needs Pebble $\geq 2.8.0$). LE staging/production (Phase 4/6) and the real VPN egress proof are OPERATOR-BLOCKED. Feature tally: **10 PASS-with-evidence, 9 PENDING, 3 OPERATOR-BLOCKED** (LE Phase 4, LE Phase 6, real VPN egress proof). Change since Rev 1: LE hermetic DNS-01 issuance flipped PENDING→PASS (the custom DNS-module-linked Caddy image is now built + proven end-to-end); LE renewal remains PENDING.